

ICON-MPC: In-Context Disturbance Estimation with Gated Delta Networks for Robust, Adaptive and Safe Quadrotor NMPC

Hakim Amer
NestAI

hakim.amer@nestai.com

Abstract

Model predictive control (MPC) for quadrotors is only as good as its disturbance estimate, and Kalman-type estimators are optimal only for linear plants, Gaussian noise and known disturbance models. Real flight breaks all three. We present ICON-MPC, a nonlinear MPC whose lumped disturbance and filtered state come from an in-context Gated Delta Network (GDN) that reads physics residuals and a Kalman-filter bank, behind a learning-free robust front end. We prove that the gated delta rule is exactly normalised-LMS system identification with forgetting, which diagonal state-space models cannot implement, and that a multi-scale low-pass training loss upper-bounds closed-loop tracking error. Keep-out constraints are tightened by an adaptive conformal margin whose long-run miss rate is bounded for any disturbance sequence. On AdaptiveQuadBench (7 disturbance families $\times$ 8 sensing/fault regimes, 1120 paired trials), ICON-MPC reduces median tracking error 4–7 $\times$ relative to the best of seven stock controllers in every regime except impacts ($p < 10^{-6}$), and is significantly better than strengthened KF and multiple-model NMPC baselines under dropout, rotor flicker and combined stress. In obstacle scenarios under combined faults it halves the unsafe-trial rate of constrained baselines (5% vs. 10–16%), with a realised conformal miss rate of 4.3–5.4% against a 5% target.

Keywords: learning-based MPC; in-context estimation; gated delta networks; adaptive conformal prediction; quadrotors; offset-free control

1. Introduction

Agile quadrotors operate under disturbances that are hard to model: wind and gusts, payload changes, degraded rotors, unmodelled aerodynamics and contact. Their sensing is also imperfect, with noisy, delayed, dropped or corrupted state estimates. Two families of controllers dominate. Adaptive controllers such as $\mathcal{L}_1$ (Hovakimyan and Cao, 2010; Wu et al., 2023) and INDI (Smeur et al., 2016; Tal and Karaman, 2021) react quickly to a lumped disturbance. However, they act on a single step and ignore constraints. Model predictive controllers (Sun et al., 2022; Hanover et al., 2022; Torrente et al., 2021; Salzmänn et al., 2023) plan over a horizon and enforce constraints, but their performance hinges on the disturbance estimate fed to the prediction model. This is the classical offset-free MPC architecture (Pannocchia and Rawlings, 2003; Maeder et al., 2009).

When can learning beat a Kalman filter? If the plant is linear, the noise Gaussian and the disturbance a known Gauss–Markov process, the Kalman filter (KF) with certainty-equivalent control is optimal (LQG separation). No learned estimator can then do better. A learned estimator can only win by exploiting violations of these assumptions. Three violations matter here: (i) the operating condition is a mixture of regimes with different statistics, so the Bayes estimator is a nonlinear, regime-weighted combination; (ii) the noise

is heavy-tailed or contains outliers and dropouts; (iii) the disturbance depends on state and input (e.g., rotor-effectiveness loss scales with thrust), so it is better treated as an unknown parameter than as a lumped signal. Here KFs, multiple-model (Magill, 1965; Blom and Bar-Shalom, 1988) and outlier-robust KFs (Ting et al., 2007) must trade one regime against another.

Approach. We keep everything that is known exactly and learn only what is not: the NMPC uses the full rigid-body model with rotor drag (Faessler et al., 2018); a learning-free front end handles stale packets, outliers and state jumps; and a sequence model performs in-context estimation of the lumped disturbance and the filtered state from physics residuals, as a correction on top of a KF bank. For the sequence model we use the Gated Delta Network (GDN) (Yang et al., 2025). We prove that its error-correcting (delta) update is normalised LMS with forgetting, the textbook recursive identification rule, which diagonal state-space models (Gu and Dao, 2023) cannot implement; we compare both empirically. For safety under state constraints, we tighten keep-out constraints by an online adaptive conformal margin (Gibbs and Candès, 2021) on the plan-prediction error. The margin carries a distribution-free long-run guarantee.

Contributions.

1. Method. ICON-MPC: a real-time (≈ 4 ms per step on one CPU thread) offset-free NMPC with an in-context GDN disturbance-and-state estimator, a robust front end and an adaptive conformal safety margin (§4).
2. Guarantees. (i) the gated delta rule is NLMS system identification with forgetting, which diagonal recurrences cannot implement; (ii) a multi-scale low-pass loss upper-bounds closed-loop tracking error; (iii) a distribution-free bound on the long-run frequency of conformal misses and hence of constraint violations (§5).
3. Evidence. A paired study on AdaptiveQuadBench (Zhang et al., 2025) (1120 trials per controller) against 7 stock and 2 strengthened baselines, plus obstacle scenarios with state constraints and ablations (§6). We report where we do not win.

2. Related Work

Adaptive and robust quadrotor control. Geometric control on $SE(3)$ (Lee et al., 2010) is the standard nominal tracker. $\mathcal{L}_1$ adaptive augmentation (Hovakimyan and Cao, 2010; Wu et al., 2023) and incremental nonlinear dynamic inversion (Smeur et al., 2016; Tal and Karaman, 2021) compensate lumped disturbances at high bandwidth. Adaptive NMPC (Hanover et al., 2022) and learning-based MPC with GP or neural dynamics (Torrente et al., 2021; Chee et al., 2022; Saviolo et al., 2022; Salzmann et al., 2023; Bauersfeld et al., 2021; Saviolo et al., 2024; Scampicchio et al., 2025), including GPs with dynamic forgetting (Amer et al., 2025b) and physics-informed models with control (Amer et al., 2025a), bring learned compensation into a predictive, constrained framework; NMPC also underpins aerial inspection (Amer et al., 2023) and dynamics emulation on multicopters (Amer and Sarabakha, 2026). Rotor drag is known to matter at speed (Faessler et al., 2018; Sun et al., 2022). AdaptiveQuadBench (Zhang et al., 2025), built on RotorPy (Folk et al.,

2023), compares these controllers; we use it unmodified and add sensing/fault regimes and obstacles.

Learning for disturbance adaptation. Neural Lander (Shi et al., 2019) and Neural-Fly (O’Connell et al., 2022) learn aerodynamic residuals and adapt a linear head online. RMA (Kumar et al., 2021) and XAdap (Zhang et al., 2024) learn an adaptation module that infers a latent from history, and XAdap is one of our baselines. Universal dynamics models (Xiao et al., 2025) and online continual learning (Sarabakha et al., 2023) adapt across platforms and tasks. Learned filters such as KalmanNet (Revach et al., 2022) and Backprop-KF (Haarnoja et al., 2016) replace or augment the KF gain. Forgione et al. (2023) frame system identification as in-context learning across a class of systems. We take the in-context view inside an offset-free MPC, with a recurrence that is provably a system-identification update and a loss tied to closed-loop error.

Sequence models. Selective state-space models (Mamba, Gu and Dao, 2023) use input-dependent diagonal recurrences. Linear transformers are fast-weight programmers (Schlag et al., 2021), and the delta rule (Widrow and Hoff, 1960) replaces their additive update with an error-correcting one. It can be trained in parallel over the sequence (Yang et al., 2024) and, with gating, gives the Gated Delta Network (Yang et al., 2025).

Safe learning-based MPC and conformal prediction. Learning-based MPC with safety guarantees typically relies on tube constructions (Mayne et al., 2005), probabilistic constraint tightening (Hewing et al., 2020a,b) or control barrier functions (Ames et al., 2019). These require a bound or a distribution on the model error. Conformal prediction gives distribution-free uncertainty sets. Its adaptive variant (ACI, Gibbs and Candès, 2021; Angelopoulos et al., 2023) retains long-run coverage under arbitrary distribution shift. It has been used to plan around uncertain other agents (Lindemann et al., 2023; Dixit et al., 2023). We instead calibrate the controller’s own plan-tracking error and use it to tighten soft state constraints (Kerrigan and Maciejowski, 2000) online.

3. Problem Setup

Dynamics. The state is $x = (p, v, q, \omega) \in \mathbb{R}^{13}$: position, velocity, unit quaternion and body rate. The input $u \in [0, f_{\max}]^4$ is the vector of rotor thrusts. We discretise with step $h = 10$ ms and write the plant as

$$x_{t+1} = f_{\theta}(x_t, u_t) + G d_t, \quad d_t = (F_t/m, J^{-1}\tau_t) \in \mathbb{R}^6. \quad (1)$$

Here f_{θ} is the rigid-body model with rotor drag (Faessler et al., 2018; Folk et al., 2023) at nominal parameters θ ; the lumped disturbance d_t , injected by G into the velocity and body-rate equations, collects everything else (wind, external wrench, payload, rotor loss, parameter error). The controller observes $y_t = \mathcal{S}_t(x_t)$, where the sensing channel $\mathcal{S}_t$ may add Gaussian or heavy-tailed noise, outliers or impacts, or return stale packets (§6.1). Given $y_{1:t}$ and $u_{1:t-1}$, the one-step physics residual

$$r_t = G^{\dagger}(y_{t+1} - f_{\theta}(y_t, u_t)) = d_t + \nu_t \quad (2)$$

is observable. Here $G^{\dagger}$ is the left inverse of G and ν_t collects measurement-noise terms; it is zero-mean under unbiased sensing.

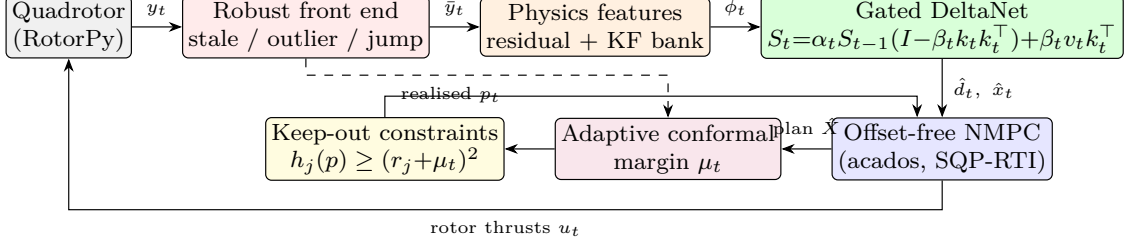


Figure 1: ICON-MPC architecture. A learning-free front end cleans the telemetry. A Gated DeltaNet reads physics features (one-step residuals and the outputs of a KF bank) and returns the lumped disturbance $\hat{d}_t$ and a filtered state $\hat{x}_t$ to an offset-free NMPC. An adaptive conformal margin μ_t , calibrated online on the realised-vs-planned position error, tightens the keep-out constraints.

Objective. Track a reference x_t^{ref} with minimal position error $\varepsilon_t = p_t - p_t^{\text{ref}}$. In obstacle scenarios, the controller must also keep the vehicle outside vertical cylinders $\mathcal{O}_j = \{p : \|p_{xy} - c_j\| \leq r_j\}$, where r_j already includes the body radius. Keep-out holds when the clearance satisfies $\min_j (\|p_{xy,t} - c_j\| - r_j) \geq 0$ for all t . We call a trial unsafe if it collides or fails to track (RMSE above 1 m or divergence).

4. Method: ICON-MPC

ICON-MPC (Fig. 1) has four layers. Each addresses one failure mode and each can be ablated (§6.4).

4.1. Offset-free nonlinear MPC

At each tick the controller solves

$$\begin{aligned}
 \min_{x_{0:N}, u_{0:N-1}, s} \quad & \sum_{k=0}^{N-1} (\|x_k - x_k^{\text{ref}}\|_Q^2 + \|u_k - u_k^{\text{ref}}\|_R^2) + \|x_N - x_N^{\text{ref}}\|_Q^2 + \sum_{k,j} (z s_{kj} + Z s_{kj}^2) \\
 \text{s.t. } \quad & x_0 = \hat{x}_t, \quad x_{k+1} = F_\theta(x_k, u_k; \hat{d}_t), \quad 0 \leq u_k \leq f_{\max}, \\
 & \|p_{xy,k} - c_j\|^2 \geq (r_j + \mu_t)^2 - s_{kj}, \quad s_{kj} \geq 0, \quad k = 1, \dots, N,
 \end{aligned} \tag{3}$$

Here F_θ integrates (1) over one shooting interval with the estimated disturbance $\hat{d}_t$ held constant over the horizon. We use a 0.5 s horizon with $N = 10$ nodes. The solver is acados (Verschuere et al., 2022) with Gauss–Newton SQP-RTI (Diehl et al., 2005), HPIPM (Frison and Diehl, 2020) and explicit Runge–Kutta integration. The keep-out constraints are soft with an exact ℓ_1 penalty (Kerrigan and Maciejowski, 2000) ($z=10^3$, $Z=10^4$), so the problem is always feasible. Whenever a feasible plan exists, the slack is zero. The input reference u_k^{ref} is the static rotor allocation of the flatness-based reference wrench minus the estimated disturbance wrench. This is an offset-free input target (Pannocchia and Rawlings, 2003; Maeder et al., 2009): under a constant disturbance and a convergent estimator, the steady-state input matches the disturbance and the tracking offset vanishes.

4.2. Learning-free robust front end

Sensor faults are handled before any learning, with three rules. Stale packets (bit-identical) are flagged, the NMPC dead-reckons $\hat{x}_t$ with its own model, and estimator kinematics are re-initialised on resume. Outliers are rejected when the innovation against the model prediction exceeds $6 \times$ a running robust scale plus a floor. Genuine jumps (e.g., impacts) are accepted once two consecutive packets agree, followed by a 10-step estimator grace window.

4.3. In-context disturbance and state estimation with Gated DeltaNet

Features and outputs. At each step the network receives a 59-dimensional feature vector ϕ_t : attitude (R_t , 9), velocity (3), body rate (3), rotor speeds (4), the previous normalised input (4), the physics residual r_t of (2) (6), kinematic-consistency increments (6), and the outputs of a bank of four lumped KFs (24) tuned for different noise/agility trade-offs. It outputs a correction to one member of the bank, $\hat{d}_t = \hat{d}_t^{\text{KF}} + \sigma_d \odot c_\vartheta(\phi_{1:t})$, together with a filtered state $\hat{x}_t$ (position, velocity, body rate and an attitude correction) and per-output log-variances. This residual parametrisation contains $c_\vartheta \equiv 0$, so in distribution the population-risk minimiser is never worse than the KF it corrects (App. A).

Gated delta recurrence. Each GDN layer (Yang et al., 2025) maps ϕ_t to a key k_t (unit norm), value v_t , query q_t , decay $\alpha_t \in (0, 1)$ and step size $\beta_t \in (0, 1)$. It updates a matrix-valued state per head:

$$S_t = \alpha_t S_{t-1} (I - \beta_t k_t k_t^\top) + \beta_t v_t k_t^\top, \quad o_t = S_t q_t. \quad (4)$$

We use two layers, $d_{\text{model}} = 64$ and four heads of dimension 16, for 51k parameters in total. By Proposition 1, (4) is normalised LMS with forgetting on $v_t \approx S k_t$, i.e. in-context system identification when keys carry disturbance regressors. For training we use the exact chunk-parallel form of (4) (Yang et al., 2024) with chunk length 64, which is $14 \times$ faster than the sequential scan. At deployment the network runs recurrently, one step per tick, on one CPU thread.

Control-aware training objective. Labels are privileged simulator quantities: the mean of the next five noise-free one-step residuals for d , and the true state for $\hat{x}$. We minimise a heteroscedastic Gaussian negative log-likelihood (Kendall and Gal, 2017) plus a multi-scale low-pass penalty on the error e_t :

$$\mathcal{L} = \sum_t \frac{1}{2} (e_t^\top \Sigma_t^{-1} e_t + \log \det \Sigma_t) + \lambda \sum_{w \in \{10, 25, 50\}} \sqrt{w} \sum_t \|\text{MA}_w(e)_t\|^2, \quad (5)$$

where MA_w is the causal moving average over w steps. By Theorem 2, such terms upper-bound closed-loop tracking error, whereas per-step MSE does not. The training data are about 10^4 five-second rollouts of KF-NMPC behaviour policies over a randomised mixture of regimes, plus one round of DAgger (Ross et al., 2011) with the learned controller in the loop. The seeds and trial indices are disjoint from the evaluation set. We train with AdamW (Loshchilov and Hutter, 2019) and a one-cycle schedule for 40 epochs, three training seeds each. A Mamba (Gu and Dao, 2023) and a GRU (Cho et al., 2014) estimator with identical features, outputs and loss serve as architecture controls.

4.4. Adaptive conformal safety margin

The NMPC predicts where the vehicle will be, but noise, faults and estimator error make the realised position deviate from the plan. Let $\hat{p}_t^{(t-\ell)}$ denote the horizontal position that the plan computed ℓ steps earlier predicted for time t (interpolated between nodes; $\ell = 10$, two node intervals). The nonconformity score is

$$e_t = \left\| p_{xy,t} - \hat{p}_t^{(t-\ell)} \right\|, \quad q_{t+1} = \max(0, q_t + \gamma (\mathbf{1}[e_t > q_t] - \alpha)), \quad \mu_t = \mu_0 + q_t. \quad (6)$$

This is the quantile-tracking form of adaptive conformal inference (Gibbs and Candès, 2021; Angelopoulos et al., 2023) with target miss rate $\alpha = 0.05$, step size $\gamma = 5$ mm and a floor $\mu_0 = 1$ cm. The margin μ_t inflates every obstacle in (3); it grows when sensing degrades (Fig. 2) and shrinks when tracking is accurate. Theorem 3 and Corollary 4 bound the long-run miss and violation frequencies for any disturbance sequence.

5. Guarantees

We give three results: one on the estimator class, one on the training objective and one on safety. Proofs are in App. A. Throughout, $\|k_t\| = 1$.

5.1. The gated delta rule is in-context system identification

Proposition 1 (GDN = NLMS with forgetting) Consider the online least-squares problem of predicting v_t from the regressor k_t with a linear map S , with instantaneous loss $\ell_t(S) = \frac{1}{2} \|v_t - Sk_t\|^2$.

- (a) The gated delta update (4) is exactly one normalised-LMS (Kaczmarz) step with step size β_t on ℓ_t , applied to the forgotten estimate $\alpha_t S_{t-1}$:

$$S_t = \alpha_t S_{t-1} - \beta_t \nabla \ell_t(\alpha_t S_{t-1}) = \alpha_t S_{t-1} + \beta_t (v_t - \alpha_t S_{t-1} k_t) k_t^\top.$$

In particular, for a linear-in-parameter disturbance channel $d_t = \varphi_t^\top \vartheta$ observed through the residual r_t , choosing $k_t = \varphi_t / \|\varphi_t\|$ and $v_t = r_t / \|\varphi_t\|$ makes the row vector S_t the normalised-LMS estimate of $\vartheta_t^\top$ with forgetting factor α_t (Widrow and Hoff, 1960).

- (b) Let the state of a recurrence be a linear embedding of S . If two keys k, k' are neither parallel nor orthogonal and $\beta, \beta' \in (0, 1]$, the transitions $A = I - \beta k k^\top$ and $A' = I - \beta' k' k'^\top$ do not commute. Hence no fixed change of coordinates makes all delta-rule transitions diagonal. Diagonal selective SSMS, whose transitions $\text{diag}(a_t)$ always commute, therefore cannot implement (4) in any linear embedding.

Rotor-effectiveness loss, mass/inertia errors and drag make d_t linear in unknown parameters with regressors from rotor speeds and velocity (part of our features). A GDN layer can thus identify them in its state, with input-dependent forgetting α_t (fast re-identification after a fault) and step size β_t (slow adaptation under noise): the two knobs a fixed-gain KF lacks.

5.2. A training loss that upper-bounds closed-loop tracking error

Let $\tilde{e}_t = d_t - \hat{d}_t$. We assume that, around the reference, the nominal NMPC closed loop is locally exponentially stable (standard with terminal ingredients or a sufficiently long horizon), hence locally input-to-state stable with respect to $\tilde{e}$ (Sontag, 2008). To first order, the tracking error is the output of a stable, linear time-invariant (LTI), multiple-input multiple-output (MIMO) filter, $\varepsilon = T(z)\tilde{e}$. Because position is a double integrator of acceleration, T is low-pass. Let $H_w(\omega) = \frac{\sin(w\omega/2)}{w\sin(\omega/2)}$ be the frequency response of MA_w , with $H_1 \equiv 1$.

Theorem 2 (Multi-scale loss bound) Let $\tilde{e}$ be wide-sense stationary and let $\lambda_1 > 0$, $\lambda_w \geq 0$. Then

$$\mathbb{E} \|\varepsilon_t\|^2 \leq c_\lambda \left(\lambda_1 \mathbb{E} \|\tilde{e}_t\|^2 + \sum_{w>1} \lambda_w \mathbb{E} \|\text{MA}_w(\tilde{e})_t\|^2 \right), \quad c_\lambda = \sup_{\omega \in [0, \pi]} \frac{\|T(e^{i\omega})\|_2^2}{\lambda_1 + \sum_{w>1} \lambda_w |H_w(\omega)|^2}. \quad (7)$$

For per-step MSE alone ($\lambda_w = 0$ for $w > 1$), $c_\lambda = \|T\|_{\mathcal{H}_\infty}^2 / \lambda_1$, attained at low frequency. Adding low-pass terms reduces c_λ by up to a factor $1 + \sum_{w>1} \lambda_w / \lambda_1$ at the frequencies where $\|T\|$ peaks.

This explains why estimators with lower per-step error can track worse: slow bias hurts tracking but is barely penalised by per-step MSE. Our loss (5) is exactly of the form (7): the likelihood term plays the role of λ_1 and $w \in \{10, 25, 50\}$ cover the closed-loop bandwidth.

5.3. Distribution-free safety from adaptive conformal margins

Theorem 3 (Long-run miss rate) Run (6) with any $q_1 \geq 0$ on any sequence of scores $e_t \in [0, B]$, which may be chosen adversarially and adaptively. Let $m_t = \mathbf{1}[e_t > q_t]$ and $\bar{Q} = \max\{q_1, B + \gamma(1 - \alpha)\}$. Then for every horizon T ,

$$\frac{1}{T} \sum_{t=1}^T m_t \leq \alpha + \frac{\bar{Q}}{\gamma T}. \quad (8)$$

Corollary 4 (Constraint-violation frequency) Suppose the plan computed at time $s = t - \ell$ has zero slack and its interpolated prediction $\hat{p}_t^{(s)}$ satisfies $\|\hat{p}_t^{(s)} - c_j\| \geq r_j + \mu_s$ for all j . Then $\min_j (\|p_{xy,t} - c_j\| - r_j) \geq \mu_s - e_t$. If in addition $\mu_0 \geq \ell\gamma(1 - \alpha)$, a collision at time t implies a miss $m_t = 1$. Consequently, the long-run fraction of time spent in collision is at most $\alpha + \bar{Q}/(\gamma T) + \rho_{\text{slack}}$, where ρ_{slack} is the fraction of plans with nonzero slack.

Theorem 3 is a long-run frequency statement, not a per-trial bound; we use $\mu_0 = 1 \text{ cm} < \ell\gamma(1 - \alpha)$ and report empirical frequencies (App. A).

6. Experiments

6.1. Setup

Benchmark. We use AdaptiveQuadBench (Zhang et al., 2025) with RotorPy (Folk et al., 2023) at 100 Hz, on 5s random trajectories and its seven disturbance families: none, wind,

Table 1: Median position RMSE [cm] per regime (140 paired trials each). Superscripts give the success rate when below 100%. Bold marks the best NMPC variant. The gain and wins compare ICON-MPC (GDN, three-seed mean) with the best stock controller in that regime, chosen post hoc; all $p < 10^{-6}$ (one-sided Wilcoxon).

Regime	Best stock controller		NMPC+	NMPC+	ICON-MPC (ours)		GDN vs. best stock	
	name	RMSE	rob. KF	MMAE	Mamba	GDN	gain	wins
Base	INDI-A	3.18	0.52	0.40	0.41	0.46	7.0×	140/140
Gauss. noise	L1-MPC	3.92	1.00	1.65	0.91	0.97	4.1×	140/140
Heavy-tail	L1-MPC	4.50	0.98	1.26	0.84	0.91	4.9×	140/140
Outliers	XAdap	6.32	1.19	1.68	1.15	1.18	5.3×	140/140
Dropout	Geo-adaptive	12.12 ^{99%}	2.31	3.60	1.77	1.83	6.6×	140/140
Rotor flicker	INDI-A	3.64	1.11	1.11	0.63	0.74	5.0×	140/140
Impacts	MPC	9.95	5.92 ^{95%}	6.00	6.52	7.04	1.4×	124/140
Stress (all)	XAdap	59.30 ^{73%}	11.06 ^{92%}	11.25 ^{96%}	10.36	10.00 ^{99%}	5.9×	138/140

external force, external torque, payload, rotor-effectiveness loss and parameter uncertainty. For each family we run 20 trials with the benchmark’s evaluation seed. Each regime therefore has 140 paired trials, in which every controller sees the same vehicle, trajectory, disturbance and noise realisation. On top of the benchmark we add seven sensing/fault regimes (details in App. B): noise (mocap/IMU-level Gaussian noise on all states and rotor speeds); heavy (Student- t , $\nu=2.5$); outliers (noise plus 2%/step 20σ glitches); dropout (noise plus frozen packets of 50–200 ms); impacts (three kicks of 0.5–1.5 m/s and 2–5 rad/s); rotor flicker (one rotor toggles between healthy and 30–60% effectiveness every 0.2–0.8 s); and stress (all of the above with aggressive trajectories).

Baselines. Stock: the seven controllers shipped with the benchmark, unmodified: geometric (Lee et al., 2010), geometric-adaptive, $\mathcal{L}_1$ -geometric (Wu et al., 2023), adaptive INDI (Smeur et al., 2016; Tal and Karaman, 2021), MPC, $\mathcal{L}_1$ -MPC and the learned XAdap (Zhang et al., 2024). Strengthened: because the stock controllers turn out to be far from ICON-MPC, we also built two strong baselines. They share everything with ICON-MPC (NMPC, rotor-drag model, front end, offset-free input target) except the estimator: a tuned outlier-robust lumped KF (Ting et al., 2007), and multiple-model adaptive estimation (MMAE) (Magill, 1965) over the same four-KF bank with forgetting. They isolate the value of the learned estimator.

Metrics and statistics. We report median position RMSE, counting a trial with RMSE > 1 m or divergence as a failure that is ranked worst. Comparisons use one-sided Wilcoxon signed-rank tests on the paired trials (Wilcoxon, 1945) and exact McNemar tests on paired unsafe outcomes (McNemar, 1947). Learned estimators are averaged per trial over three training seeds.

6.2. Tracking under disturbances and sensing faults

Against the benchmark’s controllers. Table 1 (and Fig. 3, App. C) shows that ICON-MPC reduces the median RMSE by 4.1–7.0× relative to the best stock controller in every regime

except impacts. It wins 138–140 of 140 paired trials ($p < 10^{-6}$). Under combined stress, the best stock controller (XAdap) succeeds in 73% of trials and ICON-MPC in 99%. Impacts are the exception ($1.4\times$, 124/140 wins). An impact is a state jump rather than a persistent disturbance, so little remains to be estimated, and all NMPC variants are within 20% of each other. Outside impacts, even the strengthened KF baseline is $3.3\text{--}6.1\times$ better than the best stock controller, which motivates the strengthened comparison.

Against strengthened baselines. Here only the estimator differs (Table 1; per-regime tests in Table 2, App. C). The learned estimator is significantly better where the KF assumptions break hardest: packet dropout ($1.26\times$, 117/140 wins), rotor flicker ($1.50\times$; a switching, input-dependent disturbance, the setting of Proposition 1) and combined stress ($1.11\times$, with success rising from 92.1% (KF) and 96.4% (MMAE) to 98.6%). The gains are smaller but significant under Gaussian noise, heavy-tailed noise and outliers ($1.01\text{--}1.07\times$, $p \leq 0.013$). ICON-MPC does not win in two regimes. In the nominal base regime, MMAE is better ($0.87\times$): with clean sensing a bank of well-tuned KFs is near-optimal (Fact 1 in App. A). Under impacts, the robust KF is better ($0.84\times$) because it reacts to state jumps more conservatively. Mamba matches or slightly beats GDN in most regimes (per-seed results in App. C). We thus do not claim GDN is the better drop-in sequence model; its distinguishing feature is the structure of Proposition 1.

Compute. ICON-MPC takes 3.9 ms per step on average (6.5 ms at the 99th percentile) on one CPU thread, within the 10 ms budget. Stock controllers are $10\times$ cheaper but far less accurate (Table 5, App. C).

6.3. Safety under state constraints: obstacles that intrude into the path

Scenario. We place four vertical cylinders (radius 0.1–0.3 m plus a 0.1 m body radius) at least 15 cm apart, such that the reference cuts 3–10 cm into each inflated obstacle. Tracking the reference blindly therefore guarantees a collision, and the controller must trade tracking for clearance. We test four regimes: base, noise, rotor flicker and stress (noise, outliers, dropout and flicker together), each with 140 paired trials. The stock controllers have no constraint interface and are shown for reference only. The constrained baselines use the same soft constraints (3) with either no margin, a fixed 3 cm margin, or the ACI margin (6).

Results. Constraints and margins are necessary: the stock controllers collide in essentially every trial, and the constrained KF-NMPC without a margin is unsafe in 41–73% of trials, because noise and estimation error push the realised trajectory over the boundary the plan merely touches. Without combined faults, every calibrated margin suffices: with base, noise or flicker sensing, every margin of 3 cm or ACI + 1 cm is safe in all 140 trials, and among these ICON-MPC tracks best (6.2, 8.0 and 6.3 cm) because its margin shrinks when tracking is accurate. Under combined faults, ICON-MPC is the safest: under stress, ICON-MPC is unsafe in 5.0% of trials versus 10.0–22.1% for every other constrained controller (Table 3, App. C). This is significant against all KF-based and nominal variants, including KF + ACI with the same floor (16.4%, $p = 2 \cdot 10^{-4}$), and against MMAE with a 3 cm margin ($p = 0.04$). Against MMAE + ACI + 1 cm (10.0%) the difference is not significant ($p = 0.08$), but that baseline tracks 37% worse (12.6 vs. 9.2 cm). Time in collision falls to

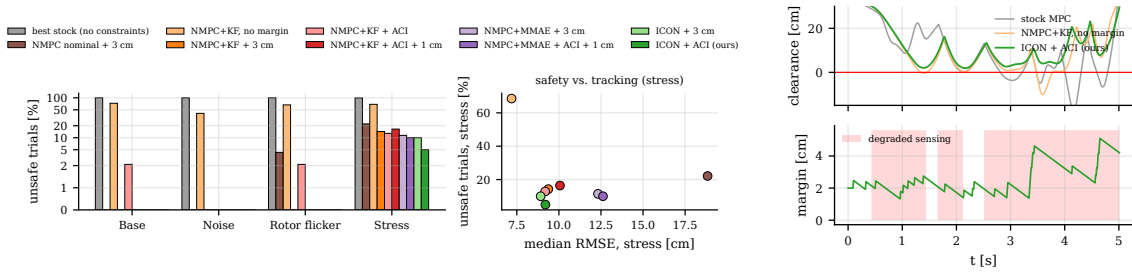


Figure 2: Left: unsafe-trial rate (collision or failure; symlog scale) and the safety–tracking trade-off under stress. Right: one stress trial (wind). Top: clearance to the nearest inflated obstacle; below the red line is a collision. Bottom: the ACI margin μ_t of ICON-MPC, which grows during degraded-sensing episodes (shaded) and decays afterwards.

0.07%, from 0.27% (MMAE + ACI) and 0.46% (KF + ACI + 1 cm). Calibration matches the theory: the realised ACI miss frequency is 4.3–5.4% in every regime and for every estimator, against the target $\alpha = 5\%$, as predicted by Theorem 3. The estimator matters because it makes the plan more predictable, so the same coverage needs a smaller margin. Figure 4 (App. C) shows a 3D rollout.

6.4. Ablations

Table 4 (App. C) shows that an NMPC without disturbance estimation (4–5 cm, even with rotor drag) is no better than the best stock controller; offset-free estimation with the robust KF alone gives 2.2–7.7 $\times$. Sequence models add on top where KF assumptions fail, and only the selective ones (Mamba, GDN) also help with clean sensing. Removing the front end is catastrophic under dropout (81% success) and stress (58%).

7. Discussion and Limitations

With clean sensing a tuned KF bank is near-optimal (Fact 1, App. A); the learned estimator pays off exactly where KF assumptions fail. All results are in simulation with our own fault models, and the theory is local (Theorem 2). The safety guarantee is a long-run frequency; a hard bound would need a tube (Mayne et al., 2005) with a certified disturbance bound, unavailable for unbounded sensor faults. GDN and Mamba perform similarly; exploiting Proposition 1 more directly is future work.

8. Conclusion

ICON-MPC couples offset-free NMPC, a robust front end, an in-context GDN estimator and an adaptive conformal margin. It tracks 4–7 $\times$ better than the benchmark’s controllers, beats strong classical estimators where their assumptions break, and halves unsafe trials under combined faults. Hardware validation is next.

References

- Abdelhakim Amer and Andriy Sarabakha. Dynamic system emulation: Fixed wing dynamics on a multicopter. *arXiv preprint arXiv:2609.06520*, 2026.
- Abdelhakim Amer, Mohit Mehndiratta, Jonas le Fevre Sejersen, Huy Xuan Pham, and Erdal Kayacan. Visual tracking nonlinear model predictive control method for autonomous wind turbine inspection. In *International Conference on Advanced Robotics (ICAR)*, pages 431–438, 2023. doi: 10.1109/ICAR58858.2023.10406329.
- Abdelhakim Amer, David Felsager, Yury Brodskiy, and Andriy Sarabakha. Modelling of underwater vehicles using physics-informed neural networks with control. In *International Joint Conference on Neural Networks (IJCNN)*, pages 1–8, 2025a. doi: 10.1109/IJCNN64981.2025.11228873.
- Abdelhakim Amer, Mohit Mehndiratta, Yury Brodskiy, and Erdal Kayacan. Empowering autonomous underwater vehicles using learning-based model predictive control with dynamic forgetting Gaussian processes. *IEEE Transactions on Control Systems Technology*, 33(5):1913–1920, 2025b. doi: 10.1109/TCST.2025.3539218.
- Aaron D. Ames, Samuel Coogan, Magnus Egerstedt, Gennaro Notomista, Koushil Sreenath, and Paulo Tabuada. Control barrier functions: Theory and applications. In *European Control Conference (ECC)*, pages 3420–3431, 2019.
- Anastasios N. Angelopoulos, Emmanuel J. Candès, and Ryan J. Tibshirani. Conformal PID control for time series prediction. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Leonard Bauersfeld, Elia Kaufmann, Philipp Foehn, Sihao Sun, and Davide Scaramuzza. NeuroBEM: Hybrid aerodynamic quadrotor model. In *Robotics: Science and Systems (RSS)*, 2021. doi: 10.15607/RSS.2021.XVII.042.
- Henk A. P. Blom and Yaakov Bar-Shalom. The interacting multiple model algorithm for systems with Markovian switching coefficients. *IEEE Transactions on Automatic Control*, 33(8):780–783, 1988.
- Nicolò Cesa-Bianchi and Gábor Lugosi. *Prediction, Learning, and Games*. Cambridge University Press, 2006.
- Kong Yao Chee, Tom Z. Jiahao, and M. Ani Hsieh. KNODE-MPC: A knowledge-based data-driven predictive control framework for aerial robots. *IEEE Robotics and Automation Letters*, 7(2):2819–2826, 2022. doi: 10.1109/LRA.2022.3144787.
- Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using RNN encoder–decoder for statistical machine translation. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1724–1734, 2014.

- Moritz Diehl, Hans Georg Bock, and Johannes P. Schlöder. A real-time iteration scheme for nonlinear optimization in optimal feedback control. *SIAM Journal on Control and Optimization*, 43(5):1714–1736, 2005.
- Anushri Dixit, Lars Lindemann, Skylar X. Wei, Matthew Cleaveland, George J. Pappas, and Joel W. Burdick. Adaptive conformal prediction for motion planning among dynamic agents. In *Learning for Dynamics and Control Conference (L4DC)*, volume 211 of *Proceedings of Machine Learning Research*, pages 300–314, 2023.
- Matthias Faessler, Antonio Franchi, and Davide Scaramuzza. Differential flatness of quadrotor dynamics subject to rotor drag for accurate tracking of high-speed trajectories. *IEEE Robotics and Automation Letters*, 3(2):620–626, 2018.
- Spencer Folk, James Paulos, and Vijay Kumar. RotorPy: A python-based multirotor simulator with aerodynamics for education and research. *arXiv preprint arXiv:2306.04485*, 2023.
- Marco Forgione, Filippo Pura, and Dario Piga. From system models to class models: An in-context learning paradigm. *IEEE Control Systems Letters*, 7:3513–3518, 2023.
- Gianluca Frison and Moritz Diehl. HPIPM: a high-performance quadratic programming framework for model predictive control. *IFAC-PapersOnLine*, 53(2):6563–6569, 2020.
- Isaac Gibbs and Emmanuel Candès. Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*, 2023.
- Tuomas Haarnoja, Anurag Ajay, Sergey Levine, and Pieter Abbeel. Backprop KF: Learning discriminative deterministic state estimators. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016.
- Drew Hanover, Philipp Foehn, Sihao Sun, Elia Kaufmann, and Davide Scaramuzza. Performance, precision, and payloads: Adaptive nonlinear MPC for quadrotors. *IEEE Robotics and Automation Letters*, 7(2):690–697, 2022.
- Lukas Hewing, Juraj Kabzan, and Melanie N. Zeilinger. Cautious model predictive control using Gaussian process regression. *IEEE Transactions on Control Systems Technology*, 28(6):2736–2743, 2020a.
- Lukas Hewing, Kim P. Wabersich, Marcel Menner, and Melanie N. Zeilinger. Learning-based model predictive control: Toward safe learning in control. *Annual Review of Control, Robotics, and Autonomous Systems*, 3:269–296, 2020b.
- Naira Hovakimyan and Chengyu Cao. $\mathcal{L}_1$ Adaptive Control Theory: Guaranteed Robustness with Fast Adaptation. *SIAM*, 2010.
- Alex Kendall and Yarin Gal. What uncertainties do we need in Bayesian deep learning for computer vision? In *Advances in Neural Information Processing Systems*, 2017.

- Eric C. Kerrigan and Jan M. Maciejowski. Soft constraints and exact penalty functions in model predictive control. In *Proc. UKACC International Conference on Control*, 2000.
- Ashish Kumar, Zipeng Fu, Deepak Pathak, and Jitendra Malik. RMA: Rapid motor adaptation for legged robots. In *Robotics: Science and Systems (RSS)*, 2021.
- Taeyoung Lee, Melvin Leok, and N. Harris McClamroch. Geometric tracking control of a quadrotor UAV on SE(3). In *IEEE Conference on Decision and Control (CDC)*, pages 5420–5425, 2010.
- Lars Lindemann, Matthew Cleaveland, Gihyun Shim, and George J. Pappas. Safe planning in dynamic environments using conformal prediction. *IEEE Robotics and Automation Letters*, 8(8):5116–5123, 2023.
- Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.
- Urban Maeder, Francesco Borrelli, and Manfred Morari. Linear offset-free model predictive control. *Automatica*, 45(10):2214–2222, 2009.
- D. T. Magill. Optimal adaptive estimation of sampled stochastic processes. *IEEE Transactions on Automatic Control*, 10(4):434–439, 1965.
- David Q. Mayne, María M. Seron, and Saša V. Raković. Robust model predictive control of constrained linear systems with bounded disturbances. *Automatica*, 41(2):219–224, 2005.
- Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947.
- Michael O’Connell, Guanya Shi, Xichen Shi, Kamyar Azizzadenesheli, Anima Anandkumar, Yisong Yue, and Soon-Jo Chung. Neural-fly enables rapid learning for agile flight in strong winds. *Science Robotics*, 7(66):eabm6597, 2022.
- Gabriele Pannocchia and James B. Rawlings. Disturbance models for offset-free model-predictive control. *AIChE Journal*, 49(2):426–437, 2003.
- Guy Revach, Nir Shlezinger, Xiaoyong Ni, Adrià López Escoriza, Ruud J. G. van Sloun, and Yonina C. Eldar. KalmanNet: Neural network aided Kalman filtering for partially known dynamics. *IEEE Transactions on Signal Processing*, 70:1532–1547, 2022.
- Stéphane Ross, Geoffrey Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 627–635, 2011.
- Tim Salzmann, Elia Kaufmann, Jon Arrizabalaga, Marco Pavone, Davide Scaramuzza, and Markus Ryll. Real-time neural MPC: Deep learning model predictive control for quadrotors and agile robotic platforms. *IEEE Robotics and Automation Letters*, 8(4):2397–2404, 2023. doi: 10.1109/LRA.2023.3246839.

- Andriy Sarabakha, Zhongzheng Qiao, Savitha Ramasamy, and Ponnuthurai Nagaratnam Suganthan. Online continual learning for control of mobile robots. In *International Joint Conference on Neural Networks (IJCNN)*, pages 1–10, 2023. doi: 10.1109/IJCNN54540.2023.10191188.
- Alessandro Saviolo, Guanrui Li, and Giuseppe Loianno. Physics-inspired temporal learning of quadrotor dynamics for accurate model predictive trajectory tracking. *IEEE Robotics and Automation Letters*, 7(4):10256–10263, 2022. doi: 10.1109/LRA.2022.3192609.
- Alessandro Saviolo, Jonathan Frey, Abhishek Rathod, Moritz Diehl, and Giuseppe Loianno. Active learning of discrete-time dynamics for uncertainty-aware model predictive control. *IEEE Transactions on Robotics*, 40:1273–1291, 2024. doi: 10.1109/TRO.2023.3339543.
- Anna Scampicchio, Elena Arcari, Amon Lahr, and Melanie N. Zeilinger. Gaussian processes for dynamics learning in model predictive control. *Annual Reviews in Control*, 60:101034, 2025.
- Imanol Schlag, Kazuki Irie, and Jürgen Schmidhuber. Linear transformers are secretly fast weight programmers. In *International Conference on Machine Learning (ICML)*, pages 9355–9366, 2021.
- Guanya Shi, Xichen Shi, Michael O’Connell, Rose Yu, Kamyar Azizzadenesheli, Animashree Anandkumar, Yisong Yue, and Soon-Jo Chung. Neural lander: Stable drone landing control using learned dynamics. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 9784–9790, 2019.
- Ewoud J. J. Smeur, Qiping Chu, and Guido C. H. E. de Croon. Adaptive incremental nonlinear dynamic inversion for attitude control of micro air vehicles. *Journal of Guidance, Control, and Dynamics*, 39(3):450–461, 2016.
- Eduardo D. Sontag. Input to state stability: Basic concepts and results. In *Nonlinear and Optimal Control Theory*, volume 1932 of *Lecture Notes in Mathematics*, pages 163–220. Springer, 2008.
- Sihao Sun, Angel Romero, Philipp Foehn, Elia Kaufmann, and Davide Scaramuzza. A comparative study of nonlinear MPC and differential-flatness-based control for quadrotor agile flight. *IEEE Transactions on Robotics*, 38(6):3357–3373, 2022.
- Ezra Tal and Sertac Karaman. Accurate tracking of aggressive quadrotor trajectories using incremental nonlinear dynamic inversion and differential flatness. *IEEE Transactions on Control Systems Technology*, 29(3):1203–1218, 2021.
- Jo-Anne Ting, Evangelos Theodorou, and Stefan Schaal. A Kalman filter for robust outlier detection. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 1514–1519, 2007.
- Guillem Torrente, Elia Kaufmann, Philipp Föhn, and Davide Scaramuzza. Data-driven MPC for quadrotors. *IEEE Robotics and Automation Letters*, 6(2):3769–3776, 2021.

- Robin Verschueren, Gianluca Frison, Dimitris Kouzoupis, Jonathan Frey, Niels van Duijken, Andrea Zanelli, Branimir Novoselnik, Thivaharan Albin, Rien Quirynen, and Moritz Diehl. *acados—a modular open-source framework for fast embedded optimal control*. *Mathematical Programming Computation*, 14(1):147–183, 2022.
- Bernard Widrow and Marcian E. Hoff. Adaptive switching circuits. In *IRE WESCON Convention Record*, pages 96–104, 1960.
- Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83, 1945.
- Zhuohuan Wu, Sheng Cheng, Pan Zhao, Aditya Gahlawat, Kasey A. Ackerman, Arun Lakshmanan, Chengyu Yang, Jiahao Yu, and Naira Hovakimyan. $\mathcal{L}_1$ -Quad: $\mathcal{L}_1$ adaptive augmentation of geometric control for agile quadrotors with performance guarantees. *arXiv preprint arXiv:2302.07208*, 2023.
- Wenli Xiao, Haoru Xue, Tony Tao, Dvij Kalaria, John M. Dolan, and Guanya Shi. AnyCar to anywhere: Learning universal dynamics model for agile and adaptive mobility. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 8819–8825, 2025.
- Songlin Yang, Bailin Wang, Yu Zhang, Yikang Shen, and Yoon Kim. Parallelizing linear transformers with the delta rule over sequence length. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- Songlin Yang, Jan Kautz, and Ali Hatamizadeh. Gated delta networks: Improving Mamba2 with delta rule. In *International Conference on Learning Representations (ICLR)*, 2025.
- Dingqi Zhang, Antonio Loquercio, Jerry Tang, Ting-Hao Wang, Jitendra Malik, and Mark W. Mueller. A learning-based quadcopter controller with extreme adaptation. *arXiv preprint arXiv:2409.12949*, 2024.
- Dingqi Zhang, Ran Tao, Sheng Cheng, Naira Hovakimyan, and Mark W. Mueller. A simulation evaluation suite for robust adaptive quadcopter control. *arXiv preprint arXiv:2510.03471*, 2025.

Appendix A. Proofs and Supporting Facts

Fact 1 (no free lunch). If the plant is linear, the process and measurement noise are Gaussian with known covariances, and d_t is a Gauss–Markov process matching the KF model, then the KF estimate is the conditional mean $\mathbb{E}[d_t | \mathcal{I}_t]$. By the separation principle, certainty-equivalent LQ control with it is optimal for the quadratic cost. Any gain from learning must therefore come from violating one of these assumptions.

Fact 2 (mixtures). Let the task distribution be a mixture over regimes $m \in \{1, \dots, M\}$, each linear-Gaussian with Kalman estimate $\hat{d}_t^{(m)}$. Then

$$\mathbb{E}[d_t | \mathcal{I}_t] = \sum_m \pi_t(m) \hat{d}_t^{(m)}, \quad \pi_t(m) = \Pr(m | \mathcal{I}_t).$$

The weights π_t are nonlinear functions of the data (likelihood ratios of the innovations). This posterior mean is the unique L_2 -optimal estimator, so whenever it is not a linear function of the data, every linear time-invariant filter, including any single KF, has strictly larger risk. MMAE implements this posterior under its own (approximate) model. A sequence model trained on samples from the mixture approximates it without requiring the regime models.

Fact 3 (residual parametrisation). With $\hat{d} = \hat{d}^{\text{KF}} + c_\vartheta$ and a class that contains $c \equiv 0$, the population-risk minimiser over the class has risk no larger than the KF. Empirical-risk minimisation adds only the usual generalisation term.

Proof of Proposition 1. (a) With $\ell_t(S) = \frac{1}{2} \|v_t - Sk_t\|^2$ we have $\nabla \ell_t(S) = -(v_t - Sk_t)k_t^\top$. Hence

$$\alpha_t S_{t-1} - \beta_t \nabla \ell_t(\alpha_t S_{t-1}) = \alpha_t S_{t-1} + \beta_t v_t k_t^\top - \alpha_t \beta_t S_{t-1} k_t k_t^\top = \alpha_t S_{t-1} (I - \beta_t k_t k_t^\top) + \beta_t v_t k_t^\top,$$

which is (4). Since $\|k_t\| = 1$, the step is normalised, and $\beta_t = 1$ projects onto $\{S : Sk_t = v_t\}$ (Kaczmarz). For a scalar channel with $r_t = \varphi_t^\top \vartheta + \nu_t$, set $k_t = \varphi_t / \|\varphi_t\|$ and $v_t = r_t / \|\varphi_t\|$. The update becomes $S_t = \alpha_t S_{t-1} + \beta_t (r_t - \alpha_t S_{t-1} \varphi_t) \varphi_t^\top / \|\varphi_t\|^2$, which is NLMS with leakage (forgetting) α_t .

(b) Write $A = I - \beta k k^\top$ and $A' = I - \beta' k' k'^\top$. Then

$$AA' = I - \beta k k^\top - \beta' k' k'^\top + \beta \beta' (k^\top k') k k'^\top, \quad A'A = I - \beta k k^\top - \beta' k' k'^\top + \beta \beta' (k^\top k') k' k^\top,$$

so $AA' - A'A = \beta \beta' (k^\top k') (k k'^\top - k' k^\top)$. This is nonzero if $k^\top k' \neq 0$ and $k \not\parallel k'$. The state transition of (4) acting on S is $S \mapsto \alpha S A$, that is, $\alpha(A^\top \otimes I)$ on $\text{vec}(S)$, which inherits the non-commutativity. If an invertible linear map P made all transitions diagonal, $P \alpha(A^\top \otimes I) P^{-1} = D_A$ and $P \alpha'(A'^\top \otimes I) P^{-1} = D_{A'}$, then the transitions would commute, because diagonal matrices commute. This is a contradiction. ■

Proof of Theorem 2. For a stable LTI MIMO system T driven by a wide-sense stationary input with spectral density matrix $S_{\tilde{e}}(\omega) \succeq 0$, the output covariance satisfies

$$\mathbb{E} \|\varepsilon_t\|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} \text{tr}(T(e^{i\omega}) S_{\tilde{e}}(\omega) T(e^{i\omega})^*) d\omega \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} \|T(e^{i\omega})\|_2^2 \text{tr} S_{\tilde{e}}(\omega) d\omega,$$

using $\text{tr}(TST^*) \leq \|T\|_2^2 \text{tr} S$ for $S \succeq 0$. Each MA_w acts componentwise with scalar response H_w , so $\mathbb{E} \|\text{MA}_w(\tilde{e})_t\|^2 = \frac{1}{2\pi} \int |H_w|^2 \text{tr} S_{\tilde{e}} d\omega$. By the definition of c_λ , we have $\|T\|_2^2 \leq c_\lambda(\lambda_1 +$

$\sum_{w>1} \lambda_w |H_w|^2$) pointwise, and integrating gives (7). The supremum is finite because $\lambda_1 > 0$. Without the λ_1 term, c_λ can be infinite: all H_w with $w \in \{10, 25, 50\}$ vanish at $\omega = 2\pi j/5$, which is why the per-step (likelihood) term is kept. ■

Proof of Theorem 3. Write $q_{t+1} = q_t + \gamma(m_t - \alpha) + c_t$ with $c_t = \max\{0, -(q_t + \gamma(m_t - \alpha))\} \geq 0$. We first show $q_t \leq \bar{Q}$ for all t , by induction. If $q_t > B \geq e_t$, then $m_t = 0$ and $q_{t+1} \leq q_t$. If $q_t \leq B$, then $q_{t+1} \leq q_t + \gamma(1 - \alpha) \leq \bar{Q}$. Summing the update over $t = 1, \dots, T$ gives $\gamma \sum_t (m_t - \alpha) = q_{T+1} - q_1 - \sum_t c_t \leq \bar{Q}$. Dividing by γT proves the claim. Conversely, if the projection is never active ($c_t \equiv 0$), then $\frac{1}{T} \sum m_t \geq \alpha - q_1/(\gamma T)$, so the miss rate converges to exactly α . ■

Proof of Corollary 4. For every j , $\|p_{xy,t} - c_j\| \geq \|\hat{p}_t^{(s)} - c_j\| - \|p_{xy,t} - \hat{p}_t^{(s)}\| \geq r_j + \mu_s - e_t$. From (6), $q_{\tau+1} - q_\tau \leq \gamma(1 - \alpha)$ for every τ , so $q_t \leq q_s + \ell\gamma(1 - \alpha)$ and $\mu_s = \mu_0 + q_s \geq q_t$ whenever $\mu_0 \geq \ell\gamma(1 - \alpha)$. A collision ($\min_j (\|p_{xy,t} - c_j\| - r_j) < 0$) then requires $e_t > \mu_s \geq q_t$, that is, $m_t = 1$. Collisions at steps whose plan has nonzero slack are counted in ρ_{slack} . ■

Scope of the safety guarantee. Theorem 3 holds for any disturbance and fault sequence, but it is a long-run frequency statement, not a hard per-trial bound. The lag condition $\mu_0 \geq \ell\gamma(1 - \alpha) = 4.75$ cm is a worst case (misses on ℓ consecutive steps). We use $\mu_0 = 1$ cm, which tracks better, and report the empirical miss and collision frequencies instead (§6.3). Inter-node chord error is a few millimetres and is absorbed by μ_0 .

A.1. Safe online aggregation

Let experts $j = 1..K$ (e.g., robust KF, fast KF, GDN) predict $\hat{d}_t^{(j)}$ before r_t is revealed, with loss $\ell_t^j = \|r_t - \hat{d}_t^{(j)}\|^2$. Because ν_t is zero-mean and independent of $\mathcal{I}_t$, $\mathbb{E}\ell_t^j = \mathbb{E}\|\hat{e}_t^{(j)}\|^2 + \mathbb{E}\|\nu_t\|^2$, so the observable loss ranks the experts by their true estimation error. On a domain with $\|r\|, \|\hat{d}\| \leq B$, the squared loss is η -exp-concave with $\eta = 1/(8B^2)$. Exponentially weighted averaging therefore guarantees, for any sequence, $\sum_t \ell_t^{\text{mix}} \leq \min_j \sum_t \ell_t^j + 8B^2 \ln K$ (Cesa-Bianchi and Lugosi, 2006). We implemented this aggregation (agg). It matched the GDN alone, for example 9.7 vs 10.0 cm median RMSE under stress, so we report the GDN in the main text.

Appendix B. Implementation Details

NMPC. State $x = (p, v, q, \omega)$ with q in $wxyz$ order; inputs are the four rotor thrusts with bounds $[0, f_{\text{max}}]$. The model is the RotorPy wrench model with rotor drag (coefficients k_d, k_z , rotor-speed floor 50 rad/s). Horizon 0.5 s with $N=10$ nodes; weights $Q = \text{diag}(10 \cdot \mathbf{1}_3, 10 \cdot \mathbf{1}_3, 0.1 \cdot \mathbf{1}_4, 0.01 \cdot \mathbf{1}_3)$ and $R = 0.01 \cdot I_4$, plus a yaw-tracking term. The solver is acados with SQP-RTI, a Gauss–Newton Hessian, FULL_CONDENSING_HPIPM and an explicit Runge–Kutta integrator. Per-stage parameters are $(m, J, r_i, \kappa_i, F_{\text{ext}}, \tau_{\text{ext}}, \text{wind})$. Obstacle slots that are not in use are placed 10 m away; very distant dummy slots degraded the QP scaling.

Estimators. The KF bank consists of four lumped random-walk KFs on $(F/m, J^{-1}\tau)$ with different measurement and process noise. The robust-KF baseline uses $r_v=r_\omega=0.15$, $q_F=10$,

$q_\tau=1$. MMAE combines the bank with posterior forgetting 0.95. The GDN has 2 layers, $d_{\text{model}}=64$, 4 heads of size 16, a short depthwise causal convolution on q, k, v , SiLU activations, ℓ_2 -normalised queries and keys, and a SiLU output gate (51k parameters). Mamba has 2 layers, $d_{\text{model}}=64$ and $d_{\text{state}}=16$ (72k parameters). The GRU has 2 layers with hidden size 64. The output head has 36 values: 18 means (6 disturbance corrections and 12 state outputs) and 18 log-variances. Training: AdamW with learning rate $2 \cdot 10^{-3}$, one-cycle schedule, batch size 32, 40 epochs, gradient clipping at 1, multi-scale weight $\lambda=1$ with windows $\{10, 25, 50\}$; validation is by held-out trial index. On one RTX 5070 GPU, the chunk-parallel form takes 19 ms per batch, against 275 ms for the sequential scan. Data: KF-NMPC behaviour rollouts (one of several KF tunings chosen per trial) on the training regime mixture, plus one DAgger round. These use benchmark seed 7 and trial indices 20–99, disjoint from evaluation (seed 42, trials 0–19).

Sensing and fault regimes. Noise standard deviations: position 5 mm, velocity 0.05 m/s, attitude 0.01 rad, body rate 0.05 rad/s and rotor speed 20 rad/s. Heavy: the same scales with Student- t ($\nu=2.5$) draws. Outliers: with probability 2% per step and per channel group, a 20σ glitch. Dropout: after 0.5 s, with probability 1% per step, the controller receives the last packet for 5–20 steps. Impacts: three kicks at uniformly random times in $[1, 4.5]$ s. Rotor flicker: after a random onset, one rotor alternates between healthy and 30–60% effectiveness every 0.2–0.8 s. Stress: noise, outliers, dropout, impacts, flicker and aggressive trajectories (position range ± 4 m, velocity and acceleration ± 6). All regime randomness uses dedicated per-trial random streams, so every controller sees identical realisations.

Obstacles. Four cylinders with radius $\sim U(0.1, 0.3)$ m plus $R_{\text{body}} = 0.1$ m. Each is placed at a random time $t_k \in [0.8, 4.7]$ s, on a random side of the reference, so that the reference penetrates the inflated cylinder by 3–10 cm. Cylinders are at least 15 cm apart, the first 0.8 s of the reference is collision-free, and placement uses up to 3000 rejection-sampling attempts. ACI: $\alpha=0.05$, $\gamma=5$ mm, $q_1=2$ cm, $\ell=10$ steps, floor $\mu_0=1$ cm.

Reproducibility. All code (controllers, regimes, training, analysis, figure and table generation) will be released. Every figure and table in this paper is produced by a single script from the raw per-trial CSV files.

Appendix C. Additional Results

Ablation and compute. Tables 4 and 5 support §6.4 and the compute paragraph of §6.2.

Training-seed variability. Table 6 lists the median RMSE of each training seed. Seed-to-seed variation (up to 15% under stress) is comparable to the gap between the two selective architectures.

Qualitative 3D rollouts. Figure 4 shows two stress trials (all faults combined). Animated versions are provided in the supplementary material.

Table 2: ICON-MPC (GDN) against the best strengthened baseline per regime (same NMPC, front end and input target; only the estimator differs). Gain = median RMSE ratio; one-sided Wilcoxon on 140 paired trials.

Regime	best strong baseline	GDN gain	wins	p (Wilcoxon)
Base	MMAE	$0.87\times$	7/140	1.00
Gauss. noise	robust KF	$1.03\times$	78/140	$4\cdot 10^{-3}$
Heavy-tail	robust KF	$1.07\times$	83/140	0.01
Outliers	robust KF	$1.01\times$	93/140	$5\cdot 10^{-5}$
Dropout	robust KF	$1.26\times$	117/140	$< 10^{-6}$
Rotor flicker	MMAE	$1.50\times$	81/140	$< 10^{-6}$
Impacts	robust KF	$0.84\times$	33/140	1.00
Stress (all)	robust KF	$1.11\times$	78/140	$3\cdot 10^{-3}$

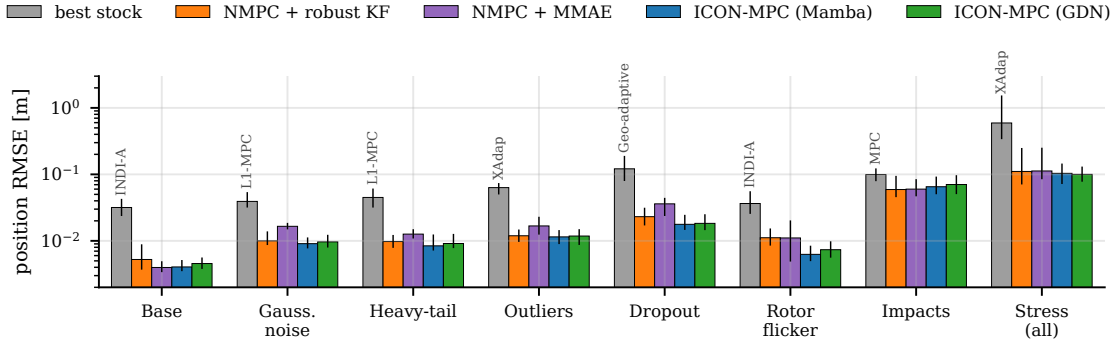


Figure 3: Median position RMSE (log scale; whiskers show the inter-quartile range) over 140 paired trials per regime. Grey: best of the seven stock controllers, whose name is shown above the bar. Orange/purple: our strengthened NMPC baselines. Blue/green: ICON-MPC with Mamba and with GDN estimators.

Table 3: Obstacle scenarios (140 paired trials each): unsafe-trial rate, median RMSE [cm] and an exact one-sided McNemar test of “baseline is less safe than ICON-MPC”.

Controller	Base (no sensing faults)			Stress		
	unsafe	RMSE	p	unsafe	RMSE	p
best stock (no constraints)	100.0%	11.8	$< 10^{-6}$	99.3%	17.3	$< 10^{-6}$
NMPC nominal + 3 cm	0.0%	10.1	1.00	22.1%	18.8	$2 \cdot 10^{-5}$
NMPC+KF, no margin	72.9%	5.2	$< 10^{-6}$	68.6%	7.2	$< 10^{-6}$
NMPC+KF + 3 cm	0.0%	7.6	1.00	14.3%	9.4	$2 \cdot 10^{-3}$
NMPC+KF + ACI	2.1%	5.6	0.12	12.9%	9.2	$6 \cdot 10^{-3}$
NMPC+KF + ACI + 1 cm	0.0%	6.4	1.00	16.4%	10.1	$2 \cdot 10^{-4}$
NMPC+MMAE + 3 cm	0.0%	7.4	1.00	11.4%	12.3	0.04
NMPC+MMAE + ACI + 1 cm	0.0%	6.3	1.00	10.0%	12.6	0.08
ICON + 3 cm	0.0%	7.5	1.00	10.0%	8.9	$8 \cdot 10^{-3}$
ICON + ACI (ours)	0.0%	6.2	–	5.0%	9.2	–

Table 4: Ablation: median RMSE [cm] (140 paired trials per regime; superscript gives the success rate when below 100%). Upper block: adding components cumulatively. Lower block: removing one component from the full model. All learned rows use training seed 0.

Variant	Base	Gauss. noise	Heavy-tail	Outliers	Dropout	Rotor flicker	Impacts	Stress (all)
NMPC, nominal model (no drag, no estimator)	5.08	4.89	4.98	5.08	6.08	9.45	9.52	22.72 ⁹⁵
+ rotor-drag model	4.02	4.01	3.98	4.01	5.02	8.52	8.83	20.98 ⁹⁴
+ robust lumped KF	0.52	1.00	0.98	1.19	2.31	1.11	5.92 ⁹⁵	11.06 ⁹²
+ learned estimator: GRU	0.52	1.02	1.00	1.23	1.87	0.79	6.53	9.97 ⁹⁸
+ learned estimator: Mamba	0.41	0.92	0.81	1.02	1.81	0.64	6.54	9.44
+ learned estimator: GDN (full)	0.46	0.97	0.91	1.19	1.79	0.74	7.53	9.34 ⁹⁸
full – robust front end	0.46	0.97	0.93	1.83	24.22 ⁸¹	0.66	6.92	55.98 ⁵⁸
full – learned state filter	0.46	0.94	0.89	1.16	1.93	0.76	7.64	9.99 ⁹⁹
full – offset-free input reference	0.46	0.97	0.91	1.19	1.80	0.75	7.53	9.53 ⁹⁹

Table 5: Controller compute per 100 Hz step (single CPU thread; 30 trials run in parallel) and outcome under stress.

Controller	mean [ms]	p99 [ms]	stress success	stress RMSE [cm]
Geometric	0.42	3.3	14%	fail
Geo-adaptive	0.50	3.4	17%	fail
L1-Geo	0.63	3.7	7%	fail
INDI-A	0.39	3.1	9%	fail
L1-MPC	0.54	5.5	10%	fail
MPC	0.32	3.3	59%	86.1
XAdap	0.62	5.2	73%	59.3
NMPC + robust KF	3.59	7.4	92%	11.1
NMPC + MMAE	4.43	9.1	96%	11.2
ICON-MPC (GDN)	3.93	6.5	98%	9.3
ICON-MPC (Mamba)	3.65	5.9	100%	9.4

Table 6: Median position RMSE [cm] per training seed (140 paired trials per regime).

Regime	GDN seed			Mamba seed		
	0	1	2	0	1	2
Base	0.46	0.47	0.45	0.41	0.41	0.41
Gauss. noise	0.97	0.98	0.94	0.92	0.89	0.91
Heavy-tail	0.91	0.92	0.93	0.81	0.82	0.91
Outliers	1.19	1.18	1.14	1.02	1.13	1.13
Dropout	1.79	1.88	1.76	1.81	1.77	1.81
Rotor flicker	0.74	0.79	0.68	0.64	0.60	0.64
Impacts	7.53	7.07	6.65	6.54	6.53	6.48
Stress (all)	9.34	9.76	10.30	9.44	9.37	10.77

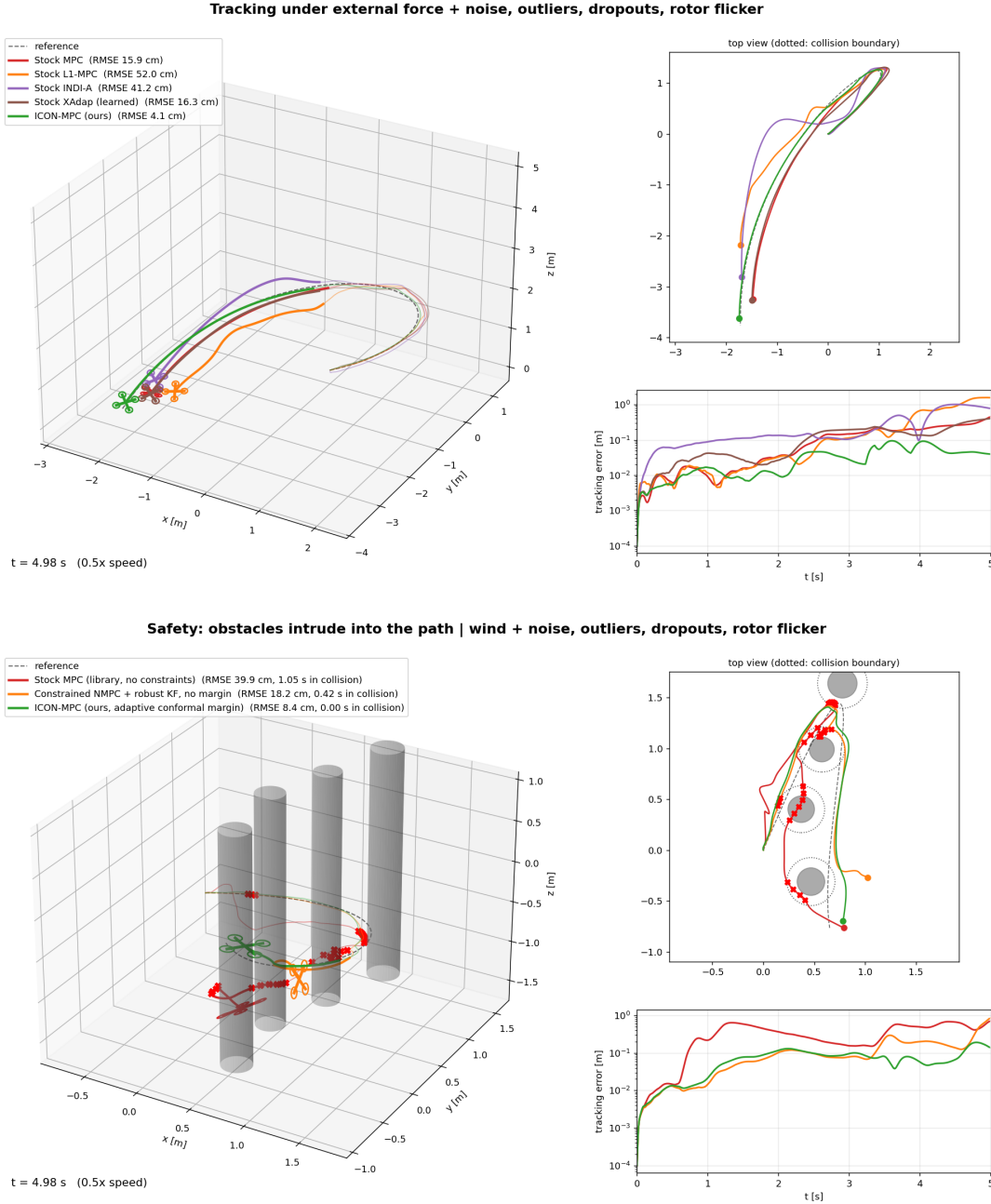


Figure 4: Top: external force plus stress. ICON-MPC (green) tracks with 4.1 cm RMSE, against 15.9–52.0 cm for the stock MPC, $\mathcal{L}_1$ -MPC, INDI-A and XAdap. Bottom: wind plus stress, with obstacles intruding into the path (red crosses mark collisions). The stock MPC spends 1.05 s in collision and the constrained KF-NMPC without a margin 0.42 s. ICON-MPC with the conformal margin never collides, with 8.4 cm RMSE.

